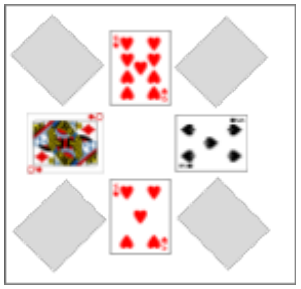
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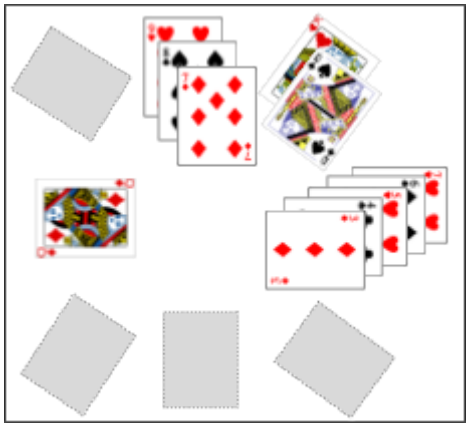
2025-2026 ● Contest 3: Kings in the Corners ● Senior Division



PROBLEM: *Kings in the Corners* is a two-person card game that uses a standard deck of 52 playing cards. The game begins by dealing 7 cards to each player alternately and placing four cards face-up into four *original* piles. There are four *corner* piles that start empty. The remaining cards form the draw pile. As the game progresses, cards are placed on piles using alternating colors and decreasing rank. For example, the 8 of Clubs (a black card) could be played on a 9 of Hearts (a red card). A King can be placed in an empty corner pile.

Play alternates between Player 1 and Player 2. Each player's turn consists of 3 parts.

1. *Move one pile:* Check if there is a pile that can be moved on top of another pile. A pile can be placed on another pile if the bottom card of the pile to be moved can be placed on the top card of the target pile (that is, it is a different color and one rank less). For example, in the images below, the pile containing 5H, 4C, 3D would be moved to the top of the pile containing 7H, 6S.


becomes


Look at the piles in clockwise order, starting with the topmost, to find one that can be moved. Look for target piles in clockwise order, starting with the topmost. Fill the empty pile with the first card from left to right in your hand with the lowest rank (regardless of suit).

2. *Play more cards:* Cards are examined from left to right in the player's hand, and the first playable card is moved according to these rules:

   - **If the card is a King,** it is played onto the first available corner pile, starting with the upper right corner and moving clockwise.

   - **If the card is not a King,** it can be played on any existing pile if it is the opposite color and exactly one rank lower than the top card of that pile, starting with the topmost pile and moving

clockwise.

Continue trying to play more cards, always looking at the cards in your hand from left to right.

3. *Draw a card:* If no pile has been moved in Step 1 and no cards have been played in Step 2, draw a single card from the Draw Pile and add it as the last card in your hand.

The first player to play all of the cards in his hand is the winner. We guarantee a winner for each game. Output a string identifying the player who won (1 or 2) followed by the top card in each pile, starting at the topmost pile and proceeding clockwise. If a pile is empty, print "E". Separate each card with a single space.

Each card is represented by a 2-character string: *rank* followed by *suit*. The ranks, from high to low, are: K (king), Q (queen), J (jack), T (ten), 9, 8, 7, 6, 5, 4, 3, 2, and A (ace). The suits are C (clubs), D (diamonds), H (hearts), and S (spades). Clubs and Spades are black; Diamonds and Hearts are red.

EXAMPLE:

Input	JC 7H AC 9S 8H 3S QD 4H KC TH 8S QH 4D 6D 6C 7S 6H 9H KS 5S
Output	1 4D 8H 6H E AC E 6C E
Explanation	This table shows each card played and the result (which is not the same as the illustrations above):

Move:	Play:	New hand:	Top card on each pile:
Player 1: Player 2:		JC AC 8H QD KC 8S 4D 7H 9S 3S 4H TH QH 6D	6C E 7S E 6H E 9H E Draw Pile: KS 5S
Player 1:	6H→7S AC	JC 8H QD KC 8S 4D	6C E **6H** E E E 9H E 6C E 6H E **AC** E 9H E
Player 1:	KC, QD, JC, 8S	JC 8H QD 8S 4D JC 8H 8S 4D 8H 8S 4D 8H 4D	6C **KC** 6H E AC E 9H E 6C **QD** 6H E AC E 9H E 6C **JC** 6H E AC E 9H E 6C JC 6H E AC E **8S** E
Player 2:	7H, TH, 9S	9S 3S 4H TH QH 6D 9S 3S 4H QH 6D 3S JH QH 6D	6C JC 6H E AC E **7H** E 6C **TH** 6H E AC E 7H E 6C **9S** 6H E AC E 7H E
Player 1:	6C→7H 4D	8H	E 9S 6H E AC E **6C** E **4D** 9S 6H E AC E 6C E
Player 1:	8H	Player 1 wins!	4D **8H** 6H E AC E 6C E

Player 1 wins the game, output the winner and the top card on each pile, starting with the topmost pile in a clockwise direction. Use an “E” for an empty pile. Output: 1 4D 8H 6H E AC E 6C E

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INPUT: Input one string representing the cards in the deck that are being used. Cards are represented as 2-character strings, each separated by a single space.

OUTPUT: Output a string identifying the player who won the game (1 or 2) and the top card played on each of the 8 piles starting in the topmost location moving clockwise, each separated by a single space.

SAMPLE INPUT:

1. JC 7H AC 9S 8H 3S QD 4H KC TH 8S QH 4D 6D 6C 7S 6H 9H KS 5S
2. TS 6H JD 5S QC 4D KH 3C 9H 7H 7D QH 7C AH 4H 8D AD JS KS 3S AS
3. JS AD 8S 2D 5D 5C 8D 5H TH 7D 4C TS QD KD JD 3H 3C 2S 7C 7H AH 7S 2H 4H KS AC QC 9H 9C 6S 4S QS 8H
4. TH JC AS AC 9C 4C JH 6H 8C 8H QC QH 3S 9H 6D 2C 5D 3D KS 4S 2H QD AH QS 3C 7C KH 7S 9D 2S 5S 6C 7D 8D 5C TS 3H 8S 2D TC JS KD 4D KC TD 9S
5. JC 7S 2D KS KC 4D TD AD 5H 2H TS 9S 5C QD AS 6S 4S AC 7H JD KH 5D 9C 8D 6H AH 8S 8H QS 3D 3H 7C QH 3C JS 2S 5S TH QC 4H 3S 6D 9D TC KD 7D
6. KC 7D 7H KS 2D AH 6D JS 7C 2H JH KD KH 2S 3C 6S AC AD 6H 4D 4S AS 5S QC TS 7S 2C 8H QH QS JC 3S 6C JD 8D 9S 9D TC 9H QD 8S 9C TH 5H 4C 8C TD 4H

SAMPLE OUTPUT:

1. 1 4D 8H 6H E AC E 6C E
2. 1 3C 9H 3S JS AD E 7D E
3. 1 2H 4S AD 7C AH E AC E
4. 2 8S AC 5C AH 2S QC AS QD
5. 2 5D AS AD AC 3H 7H AH E
6. 1 AC AS AH 4S AD 9C 6H 6C

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TEST DATA

TEST INPUT:

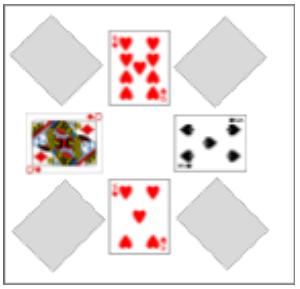
1. 4S QH 2H 9D KS 8H 2C KC 5H 7D 3H 9C QD 5S AD 6C 9S 7H 4D 8S QC AS
2. 6C 9C KH TC 9H 2H JS 2S KC 9S 5H 6D AH QH TD 7D QD TS AC KS 4H 8D TH 7C 8C 4C
3. 5S 3H 4D KH 7S JD JH 8H 2H 7H 2S 6D AH QC 7C 8S 5C 4H AC JS 4C 9H 6H TC TH 6C 3S JC QD 9D
4. QS 9H AC QD TC 9C 6C 7S AS AD QC 7C 8H KH 2S 8S 3D JD JS QH AH 5H 6S 4S KC 2D 8D 4C 4D 2H KD 3H 5C 5D 6D 7H TD 7D JH 6H JC 3S 3C 5S 4H KS 9D TS 2C
5. 3C KD 2C 5C QC 4H QS 6S 5D 4S TH JH 6C KH 7S JS 2H 9C JC 7H 2S 8C AD KS AC 5S AH 8S TS 3H 6H 6D 7C 7D AS 9S 3D QH 3S TC 4C 4D KC 9H 5H 2D JD 8D TD 9D 8H
6. TH 3S 7D 9H 9D 2C TS 8H 6C 3H 2D 4C AD 2S AC 8S 5C JH TC QS 8C 4S QC JS 9C AH 6H 5H 9S QD 5D KD 3D KC 4D 8D 7C 5S 3C JD 6S KS 7S

TEST OUTPUT:

1. 1 AD QD 2H E 9S E 2C E
2. 1 AC KH 5H QH AH 9C 9H E
3. 2 AC 6C 4D E 8H E AH E
4. 2 AH 3H AD AS QH JH AC E
5. 2 AC AH AS TC 8S JC AD KC
6. 1 8C AH AC 8H QS E AD E

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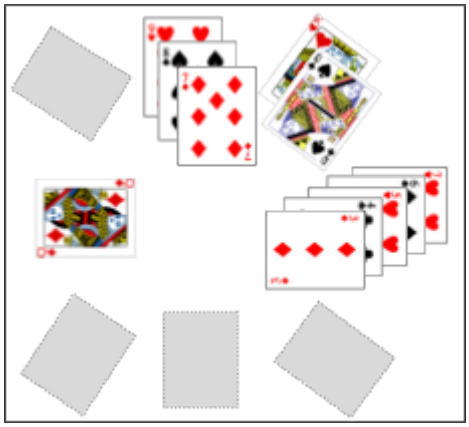
问题：*角落国王* 是一款双人纸牌游戏，使用一副标准的 52 张扑克牌。游戏开始时，每名玩家轮流发牌，每人各7张，并面朝上放置四张牌，形成四个*初始牌堆*。另有四个*角落牌堆* 初始为空。剩余的牌组成抽牌堆。在游戏过程中，玩家需按照颜色交替、点数递减的规则将牌放到牌堆上。例如，梅花 8（黑色牌）可以放在红桃 9（红色牌）的上面。而 K 只能放在空的角落牌堆中。

玩家1和玩家2轮流出牌。每位玩家的回合均包含三个环节：

1. *移动整个牌堆：*检查是否有某个牌堆可以整体移动到另一个牌堆的上面。
一个牌堆可以移动到另一个牌堆上的条件是：待移动牌堆的最底下一张牌，能够放到目标牌堆的最上面一张牌之上（即颜色不同且点数小1）。例如，在下图中，包含“ 红桃 5、梅花 4、方块 3 ”的牌堆可以移动到包含“ 红桃 7、黑桃 6 ”的牌堆之上。



变为



按顺时针顺序检查各个牌堆，从最上方的牌堆开始，寻找可以移动的牌堆。寻找目标牌堆时，同样按顺时针顺序，从最上方的牌堆开始检查。如果某个牌堆空了，则用手中从左往右数，点数最小（无论花色）的第一张牌将其填充。

2. *打出多张牌：*玩家按从左到右的顺序检查自己手中的牌，将第一张符合条件的牌按以下规则打出：

   - **如果这张牌是 K**，则将其放入第一个可用的角落牌堆。角落牌堆的填充顺序是从右上角开始，按顺时针方向进行。

   - **如果这张牌不是 K**, 则可以将其放入任何一个现有牌堆，条件是这张牌的颜色与该牌堆顶牌的颜色相反，且牌面点数正好小1。检查牌堆时，从最上方的牌堆开始，按顺时针顺序进行。

继续尝试打出更多的牌，始终从左到右检查手中的牌。

3. *抽一张牌：*如果在步骤 1 中没有移动任何牌堆，并且在步骤 2 中没有打出任何牌，则从抽牌堆中抽一张牌，将其作为手牌中的最后一张牌。

先将手中所有牌都出完的玩家获胜。题目保证每局游戏都会产生一名胜者。输出一行字符串，先输出获胜玩家的编号（1 或 2），然后从最上方的牌堆开始，按顺时针顺序依次输出 8 个牌堆的顶牌。如果某个牌堆为空，则输出“ E ”。每张牌之间用一个空格分隔。

每张牌用两个字符的字符串表示：*点数*后跟*花色*。点数从高到低依次为：K（国王）、Q（皇后）、J（杰克）、T（10）、9、8、7、6、5、4、3、2、A 牌。花色分别为：C（梅花）、D（方块）、H（红桃）、S（黑桃）。梅花和黑桃为黑色；方块和红桃为红色。

示例:

输入	JC 7H AC 9S 8H 3S QD 4H KC TH 8S QH 4D 6D 6C 7S 6H 9H KS 5S
输出	1 4D 8H 6H E AC E 6C E

解释说明

该表格展示了每张打出的牌及其结果（与上述图示结果不同）：

移动:	出牌:	新的手牌:	**各牌堆顶牌：**
玩家1: 玩家2:		JC AC 8H QD KC 8S 4D 7H 9S 3S 4H TH QH 6D	6C E 7S E 6H E 9H E 抽牌堆: KS 5S
玩家1:	6H→7S AC	JC 8H QD KC 8S 4D	6C E **6H** E E E 9H E 6C E 6H E **AC** E 9H E
玩家1:	KC, QD, JC, 8S	JC 8H QD 8S 4D JC 8H 8S 4D 8H 8S 4D 8H 4D	6C **KC** 6H E AC E 9H E 6C **QD** 6H E AC E 9H E 6C **JC** 6H E AC E 9H E 6C JC 6H E AC E **8S** E
玩家2:	7H, TH, 9S	9S 3S 4H TH QH 6D 9S 3S 4H QH 6D 3S JH QH 6D	6C JC 6H E AC E **7H** E 6C **TH** 6H E AC E 7H E 6C **9S** 6H E AC E 7H E
玩家1:	6C→7H 4D	8H	E 9S 6H E AC E **6C** E **4D** 9S 6H E AC E 6C E
玩家1:	8H	Player 1 wins!	4D **8H** 6H E AC E 6C E

玩家 1 获胜，请输出获胜者编号以及每个牌堆的顶牌，从最上方的牌堆开始按顺时针顺序排列。空牌堆用“ E ”表示。输出：1 4D 8H 6HE AC E 6C E

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输入: 输入一行字符串，代表游戏中所用的牌堆顺序。每张牌用 2 个字符的字符串表示，每张牌之间用一个空格分隔。

输出: 输出一行字符串，先输出获胜玩家的编号（1 或 2），然后按顺时针顺序，从最上方的牌堆开始，依次输出 8 个牌堆的顶牌，每项之间用一个空格分隔。

样例输入：

1. JC 7H AC 9S 8H 3S QD 4H KC TH 8S QH 4D 6D 6C 7S 6H 9H KS 5S
2. TS 6H JD 5S QC 4D KH 3C 9H 7H 7D QH 7C AH 4H 8D AD JS KS 3S AS
3. JS AD 8S 2D 5D 5C 8D 5H TH 7D 4C TS QD KD JD 3H 3C 2S 7C 7H AH 7S 2H 4H KS AC QC 9H 9C 6S 4S QS 8H
4. TH JC AS AC 9C 4C JH 6H 8C 8H QC QH 3S 9H 6D 2C 5D 3D KS 4S 2H QD AH QS 3C 7C KH 7S 9D 2S 5S 6C 7D 8D 5C TS 3H 8S 2D TC JS KD 4D KC TD 9S
5. JC 7S 2D KS KC 4D TD AD 5H 2H TS 9S 5C QD AS 6S 4S AC 7H JD KH 5D 9C 8D 6H AH 8S 8H QS 3D 3H 7C QH 3C JS 2S 5S TH QC 4H 3S 6D 9D TC KD 7D
6. KC 7D 7H KS 2D AH 6D JS 7C 2H JH KD KH 2S 3C 6S AC AD 6H 4D 4S AS 5S QC TS 7S 2C 8H QH QS JC 3S 6C JD 8D 9S 9D TC 9H QD 8S 9C TH 5H 4C 8C TD 4H

样例输出：

1. 1 4D 8H 6H E AC E 6C E
2. 1 3C 9H 3S JS AD E 7D E
3. 1 2H 4S AD 7C AH E AC E
4. 2 8S AC 5C AH 2S QC AS QD
5. 2 5D AS AD AC 3H 7H AH E
6. 1 AC AS AH 4S AD 9C 6H 6C

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测试数据

测试输入：

1. 4S QH 2H 9D KS 8H 2C KC 5H 7D 3H 9C QD 5S AD 6C 9S 7H 4D 8S QC AS
2. 6C 9C KH TC 9H 2H JS 2S KC 9S 5H 6D AH QH TD 7D QD TS AC KS 4H 8D TH 7C 8C 4C
3. 5S 3H 4D KH 7S JD JH 8H 2H 7H 2S 6D AH QC 7C 8S 5C 4H AC JS 4C 9H 6H TC TH 6C 3S JC QD 9D
4. QS 9H AC QD TC 9C 6C 7S AS AD QC 7C 8H KH 2S 8S 3D JD JS QH AH 5H 6S 4S KC 2D 8D 4C 4D 2H KD 3H 5C 5D 6D 7H TD 7D JH 6H JC 3S 3C 5S 4H KS 9D TS 2C
5. 3C KD 2C 5C QC 4H QS 6S 5D 4S TH JH 6C KH 7S JS 2H 9C JC 7H 2S 8C AD KS AC 5S AH 8S TS 3H 6H 6D 7C 7D AS 9S 3D QH 3S TC 4C 4D KC 9H 5H 2D JD 8D TD 9D 8H
6. TH 3S 7D 9H 9D 2C TS 8H 6C 3H 2D 4C AD 2S AC 8S 5C JH TC QS 8C 4S QC JS 9C AH 6H 5H 9S QD 5D KD 3D KC 4D 8D 7C 5S 3C JD 6S KS 7S

测试输出：

1. 1 AD QD 2H E 9S E 2C E
2. 1 AC KH 5H QH AH 9C 9H E
3. 2 AC 6C 4D E 8H E AH E
4. 2 AH 3H AD AS QH JH AC E
5. 2 AC AH AS TC 8S JC AD KC
6. 1 8C AH AC 8H QS E AD E